Lecture 4b: Application of the Expected Utility Hypothesis. The Markowitz Portfolio Allocation Problem for Risky Assets

CHEME 5760 Cornell University

Objectives. In this lecture, we will:

- **O1**: Introduce the Markowitz portfolio problem in words: Markowitz portfolio optimization is a basic quantitative finance concept: maximize reward for a specified risk -or- minimum risk for a specified reward

- **O2**: Introduce the reward (return) and risk (variance) for a portfolio of risky assets, e.g., equities (stocks or exchange-traded funds, ETFs). Define the efficient frontier.

- **O3**: Compute the efficient frontier for a portfolio of risky assets without short-selling (no stock borrowing allowed).

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low-risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Jupyter Notebooks

Repository: https://github.com/varnerlab/CHEME-5760-Examples-F23.git

- **Example**: CHEME-5760-L4b-PortfolioAllocation.ipynb

- **Rational**: Investors are rational and risk-averse.
  - When presented with two portfolios that offer the same expected benefit, a risk-averse investor will always choose the less risky portfolio.
  - When presented with two portfolios that offer the same risk, a rational investor will always choose the portfolio with the largest reward.
  - If an investor wants a higher reward, they must accept more risk.
- **Markowitz**: for a set of assets $\mathcal{A}$ composed of risky assets, e.g., equities or equity derivatives, and risk-free assets, e.g., treasury debt securities, the Markowitz portfolio allocation problem estimates the weight ω_a of asset a in portfolio $\mathcal{P}$ for $\forall a \in \mathcal{A}$ that **minimizes the risk for a user-specified return**.

O2: Introduce the reward (return) and risk (variance) for a portfolio of risky assets, e.g., equities (stocks or exchange-traded funds, ETFs).

We have assets set $\mathcal{A}$ which can be arranged into a portfolio $\mathcal{P}$. The return of asset i in portfolio $\mathcal{P}$ on some time basis $\mathcal{B}$, e.g., days, weeks, etc is denoted as r_i:

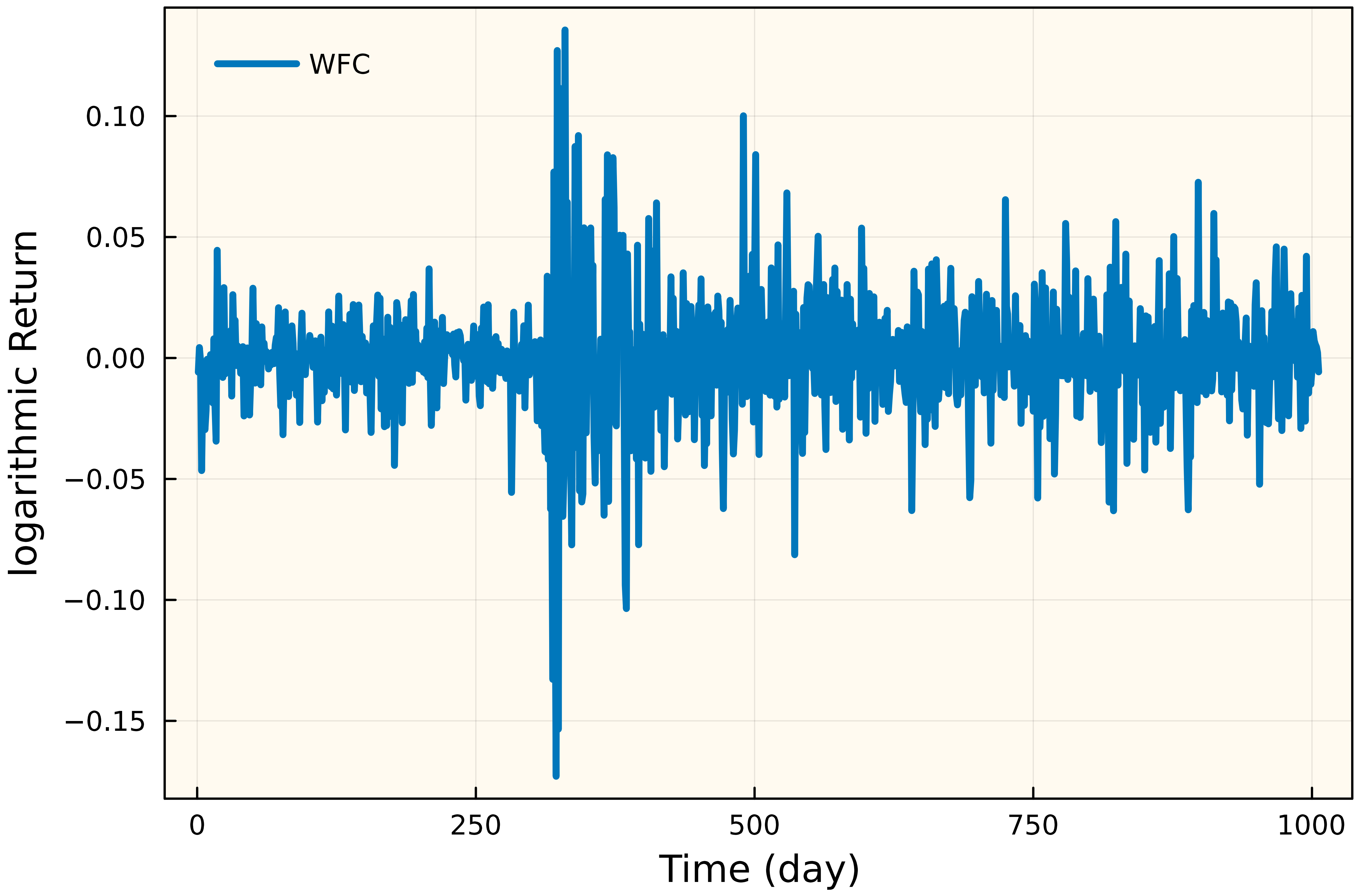
$$r_i \equiv \log\left(\frac{P_i^t}{P_i^{t-1}}\right)$$

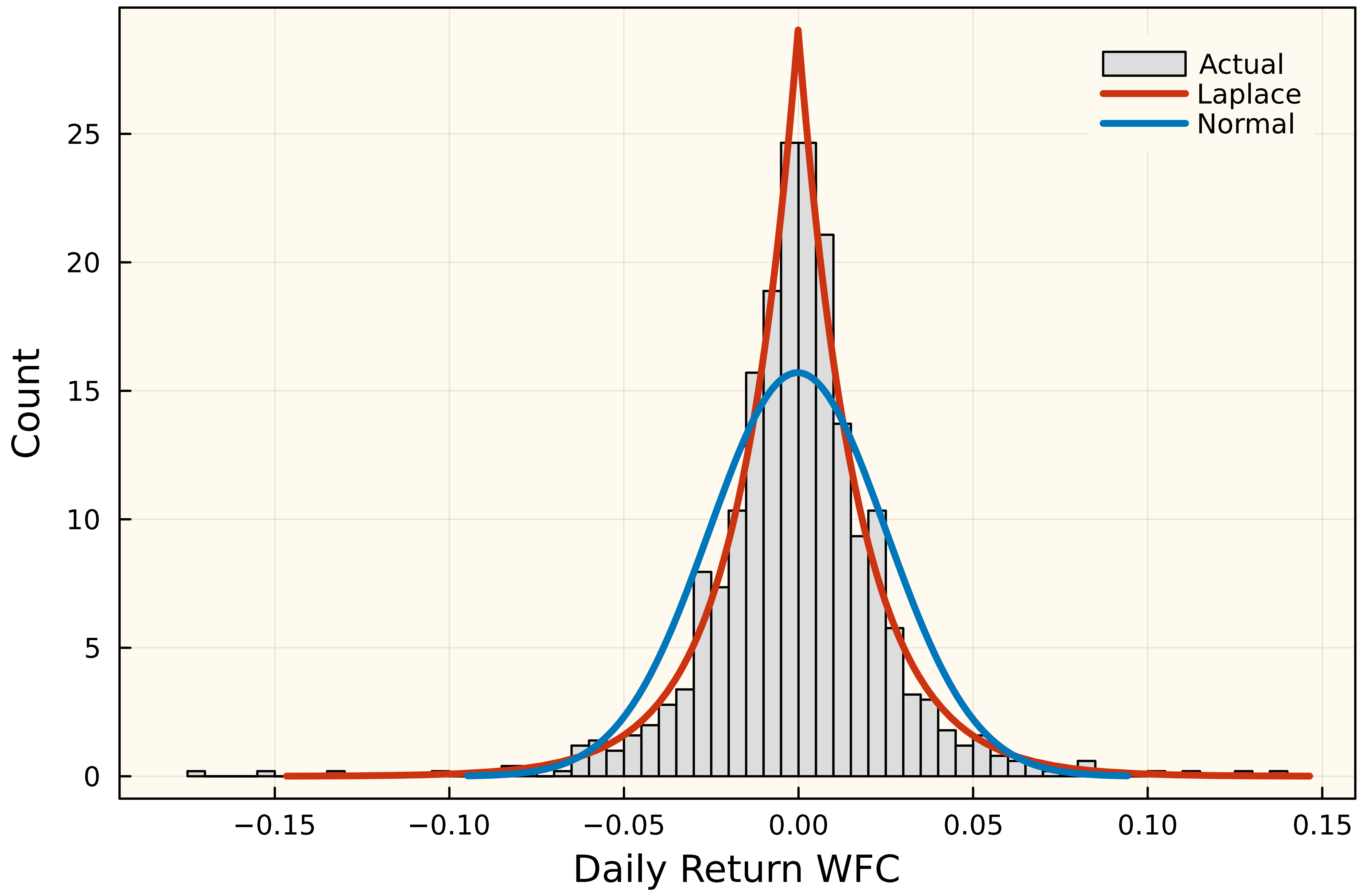
where P_i^t is the price of asset i at time t. Then, the expected return of portfolio $\mathcal{P}$ denoted by $\mathbb{E}\left(r_{\mathcal{P}}\right)$ is:

$$\mathbb{E}\left(r_{\mathcal{P}}\right) = \sum_{i\in\mathcal{P}} \omega_i \cdot \mathbb{E}\left(r_i\right)$$

where w_i denotes the fraction of asset i in the portfolio $\mathcal{P}$, and $\mathbb{E}\left(r_i\right)$ denotes the expected return of asset i on time basis $\mathcal{B}$. Finally, the fractions w_i must sum to one: $\sum_{i\in\mathcal{P}} \omega_i = 1$.

Logarithmic returns for risky assets may not always follow a Normal distribution, as often assumed. For instance, the WFC return seems to follow a Laplace distribution.





The return of asset i in portfolio $\mathcal{P}$ for some time basis $\mathcal{B}$ is given by r_i. Then, the variance of a portfolio $\mathcal{P}$ is given by:

$$\sigma_{\mathcal{P}}^2 = \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{P}} w_i w_j \text{cov}\left(r_i, r_j\right)$$

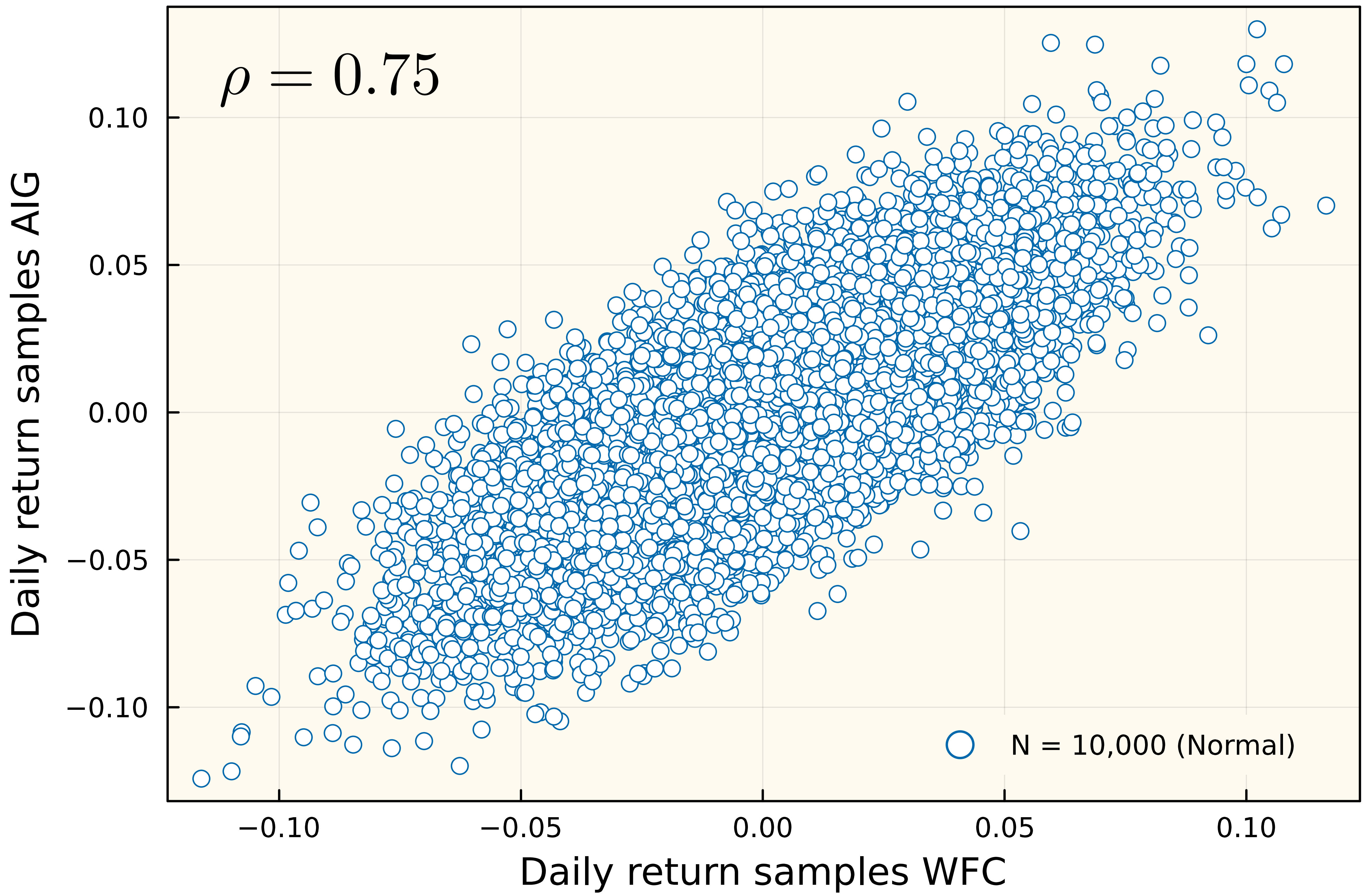
where $w_\star$ denotes the fraction of asset $\star$ in the portfolio $\mathcal{P}$, and $\text{cov}\left(r_i, r_j\right)$ denotes the covariance between the return on assets i and j. The $\text{cov}\left(r_i, r_j\right)$, which quantifies the relationship between assets i and j, is given by:

$$\text{cov}\left(r_i, r_j\right) = \sigma_i \sigma_j \rho_{ij}$$

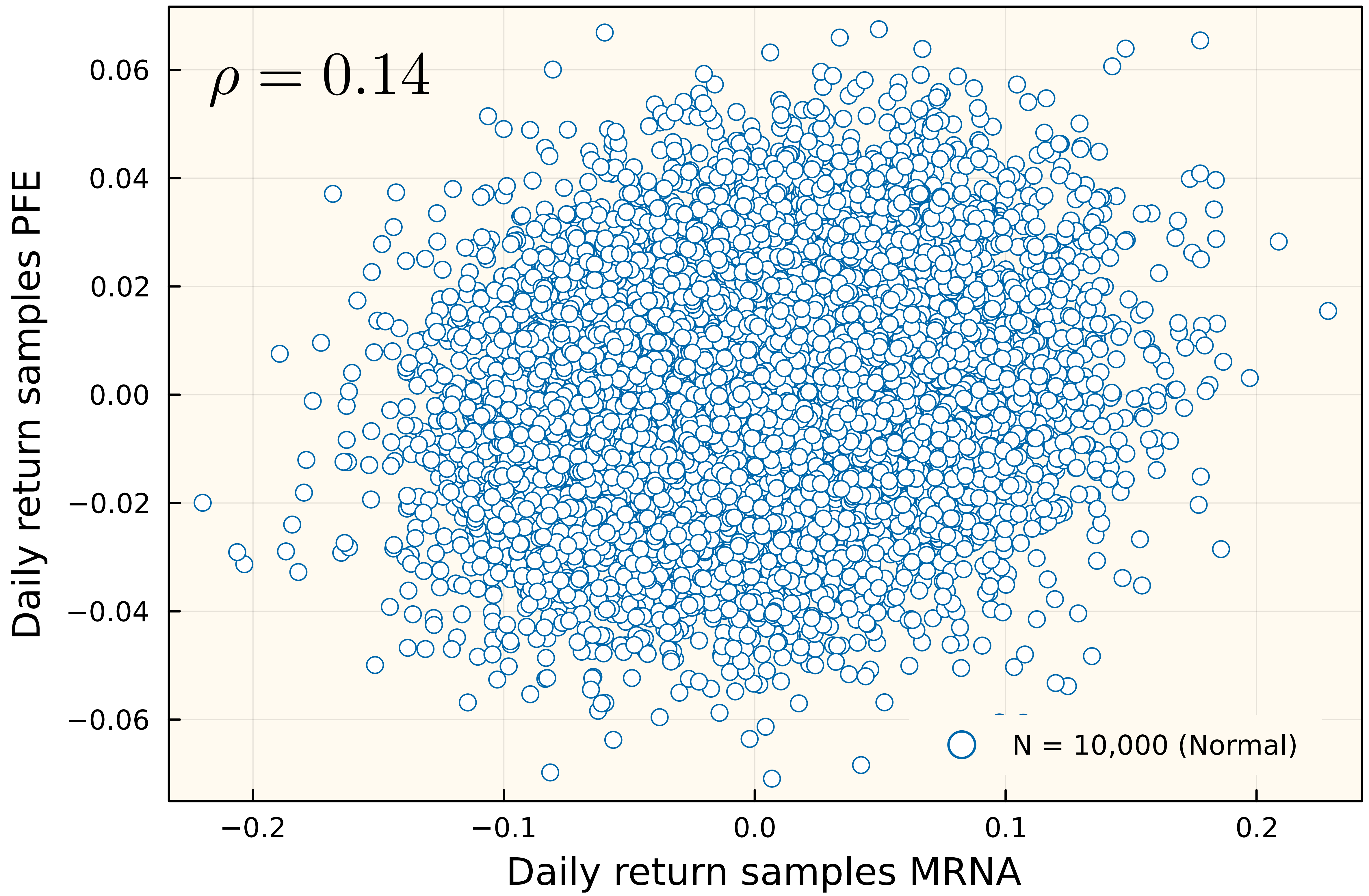
where $\sigma_\star$ denote the standard deviation of the return of asset $\star$, and ρ_{ij} denotes the correlation between assets i and j.

$\mathrm{cov}(x_1, x_2) < 0$ $\qquad$ $\mathrm{cov}(x_1, x_2) = 0$ $\qquad$ $\mathrm{cov}(x_1, x_2) > 0$

Covariance measures how two assets move in relation to each other. Ticker symbols tend to have correlated returns, but the patterns are not always predictable.



High-correlation (AIG and WFC are both financials)

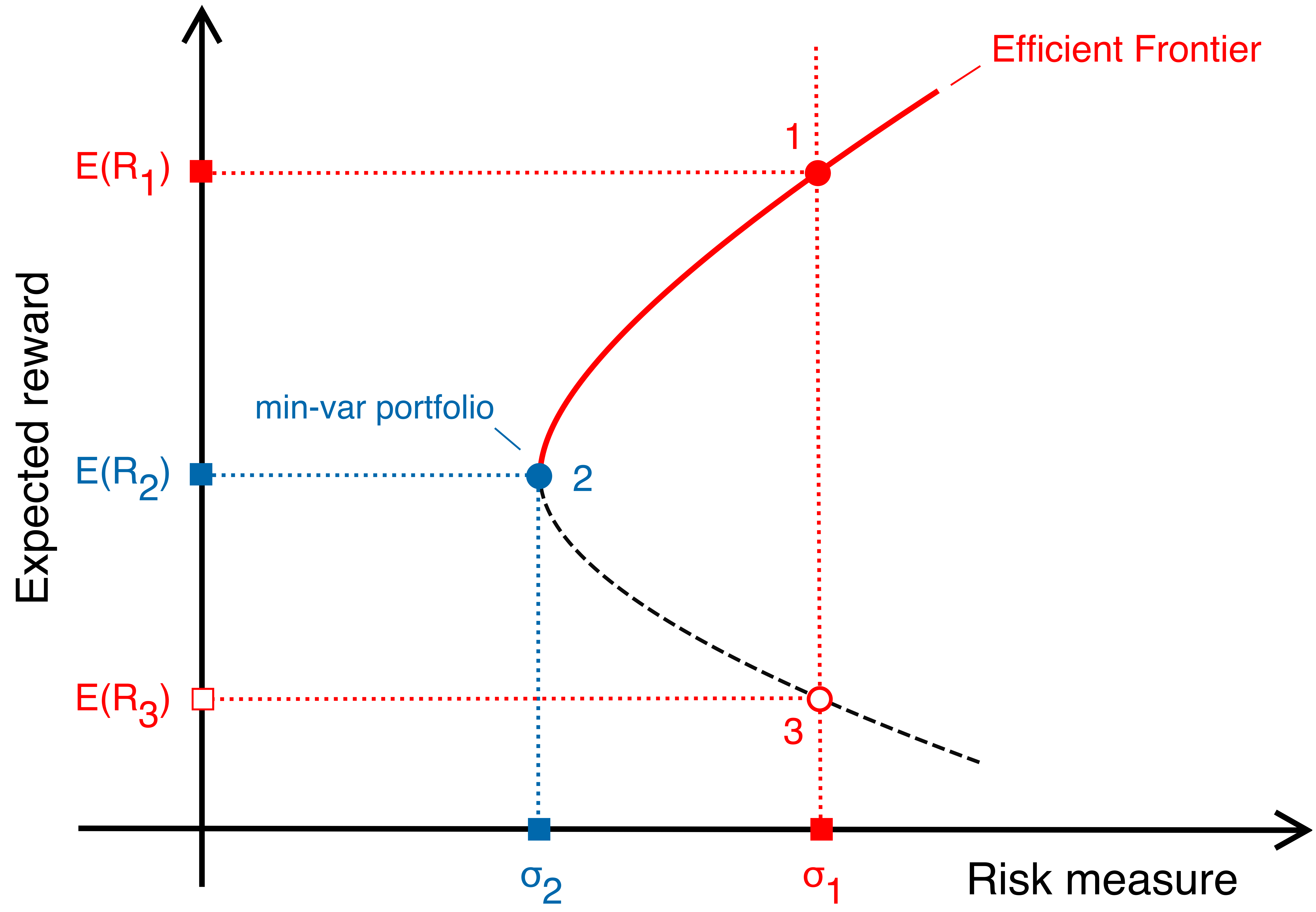


Low-**correlation** (MRNA and PFE are both healthcare)

The Markowitz allocation problem for a portfolio $\mathcal{P}$ composed of only risky assets is given by:

$$
\begin{aligned}
\text{minimize } \sigma_{\mathcal{P}}^2 &= \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{P}} w_i w_j \operatorname{cov}(r_i, r_j) \\
\text{subject to } \mathbb{E}(r_{\mathcal{P}}) &= \sum_{i \in \mathcal{P}} w_i \cdot \mathbb{E}(r_i) \geq R^* \\
\sum_{i \in \mathcal{P}} \omega_i &= 1 \\
\text{and } w_i &\geq 0 \qquad \forall i \in \mathcal{P}
\end{aligned}
$$

The term R^* is the minimal required return for $\mathcal{P}$, and 1^T is a vector of ones. The $w_i \geq 0 \ \forall i \in \mathcal{P}$ constraint forbids short selling (borrowing shares); If short selling is allowed, this constraint can be removed.


Efficient Frontier
1
E(R1)
Expected reward
min-var portfolio
E(R2)
2
E(R3)
3
σ2
σ1
Risk measure

Today?

- We introduced the Markowitz portfolio problem in words and symbols: Markowitz portfolio maximizes reward for a specified risk -or- minimizes risk for a specified reward
- We introduced typical reward (return) and risk (variance) models for a portfolio of risky assets, e.g., equities (stocks or exchange-traded funds, ETFs), and defined the efficient frontier
- We computed the efficient frontier for a portfolio of risky assets without short-selling (no stock borrowing).